

Foundation Certificate in Higher Education
DOC325 Digital Circuits and Logic Design
Model Exam Paper

1. Find the following number conversions with steps.

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| a. 1111011010101011_2 to Hexadecimal | d. 3356 to Hexadecimal |
| b. 1101010_2 to Decimal | e. $A5F6_{16}$ to Binary |
| c. 289 to Binary | f. 270_{16} to Decimal |

2. Perform the following operation between the binary numbers. (2x2 marks)

- a. $10101010 + 11001100 + 111000$
b. $1111000110 - 110101101 - 11011$

3. Simplify the following Boolean expression using Boolean Algebra Laws. (14 marks)

$$(A+C)(AB+A\bar{B})+AC+C$$

4. Find the simplified expression for the function $f(A,B,C,D) = \Sigma (0,2,5,7,10,11,12,13,14,15)$ using K-Maps. Clearly show the necessary steps. (8 marks)

5. Design a logic circuit that controls the alarm buzzer for an elevator in a building. The alarm will be turned on when the weight exceeds 900 kg or there was a power cut and the elevator is moving, or someone pressed the emergency button, and the elevator is moving.

Assume,

- A is a weight sensor in the elevator.
- B is a signal which indicates whether the elevator is moving.
- C is the power cut signal.
- D is an emergency signal.
- O is the signal for the alarm buzzer.

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| i. Construct the truth table and write the minterms. | (12 marks) |
| ii. Write the logic expression in SOP form. | (11 marks) |
| iii. Obtain the minimized logic expression using the K-map. | (9 marks) |
| iv. Implement the minimized expression using logic gates. | (5 marks) |

6. In digital electronics, both combinational and sequential circuits are the most widely used circuits.

- i. Give 3 main objectives in designing logic circuits. (3 marks)
- ii. State 3 differences between the Combinational and Sequential circuits and give one example each. (5 marks)
- iii. Explain how the priority encoder differs from binary encoder. (2 marks)
- iv. Draw the Truth Table for the **8x3 priority encoder**. (8 marks)
- v. Explain how to implement a full adder using half adders with a diagram. Clearly show the intermediate logic expressions. (7 marks)